IGCSE 0607 Practice Questions — E7.1 — Combined Questions & Markscheme

Pythagoras' theorem

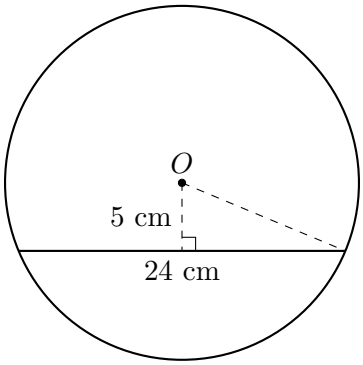
Total Questions: 6 — **Total Marks:** 38 Marks

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- **E7.1** Pythagoras' theorem
  → **Paper 2 (Non-Calculator)** *(4 Qs)* · **Paper 4 (Calculator)** *(2 Qs)*

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1 A chord of a circle, centre O, has length 24 cm. The perpendicular distance from the centre of the circle to the chord is 5 cm.



Calculate the radius of the circle.

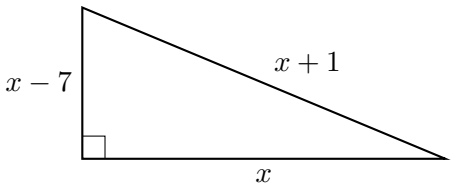
..... cm **[3]**

2 The point A has coordinates $(1, 4)$ and the point B has coordinates $(k, 19)$.
The exact distance between point A and point B is 17.

Find the two possible values of k.

$k =$ ..... , $k =$ ..... **[4]**

3 The diagram shows a right-angled triangle.
All lengths are in centimetres.



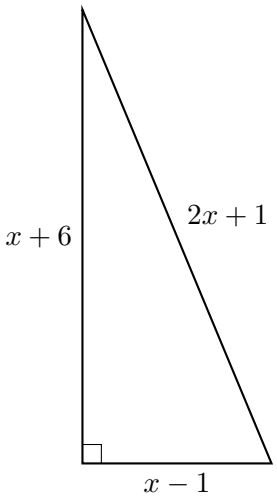
(a) Show that $x^2 - 16x + 48 = 0$.

[3]

(b) Solve the equation to find the value of x, and hence calculate the perimeter of the triangle.

Perimeter = ... cm **[4]**

4 The diagram shows a right-angled triangle.
The lengths of the sides are $x - 1$, $x + 6$, and $2x + 1$, where x is an integer.
All lengths are in centimetres.



(a) Show that $x^2 - 3x - 18 = 0$.

[4]

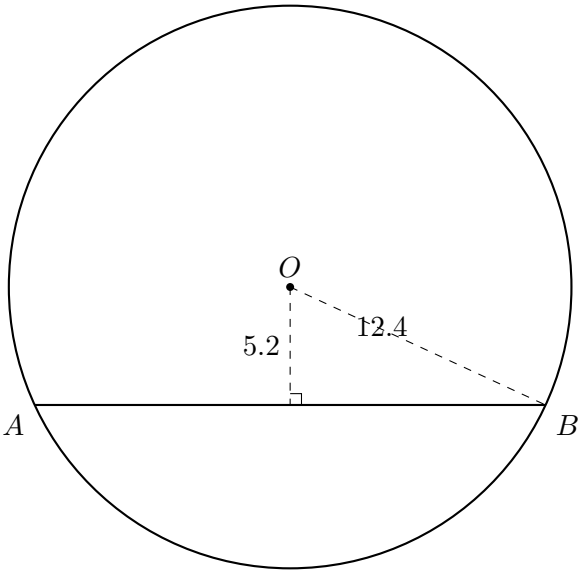
(b) Solve the equation by factorising to find the value of x.
You must show your working.

$x =$ ..... **[3]**

(c) Work out the area of the triangle.

..... cm^2 **[3]**

5 The diagram shows a circle, centre O, with radius 12.4 cm.
AB is a chord of the circle.
The perpendicular distance from O to the chord AB is 5.2 cm.



(a) Calculate the length of the chord AB.

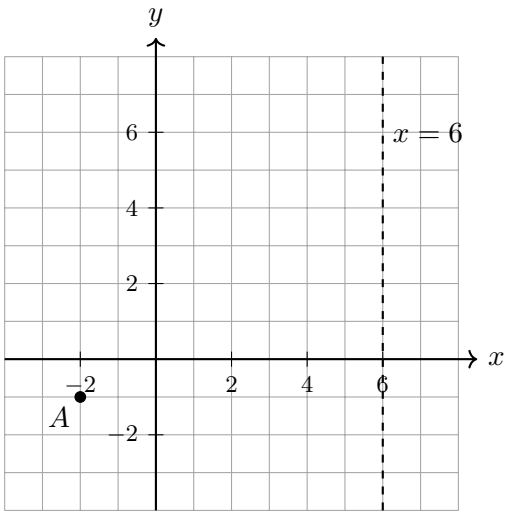
..... cm **[3]**

(b) Another chord CD is drawn parallel to AB and on the opposite side of the centre O.
The length of the chord CD is 18 cm.

Calculate the perpendicular distance between the two chords AB and CD.

..... cm **[4]**

6 The diagram shows a coordinate grid.
Point A is $(-2, -1)$ and point B lies on the line $x = 6$.
The y-coordinate of B is positive.
The distance AB is 10 units.



(a) Find the y-coordinate of point B.

$y =$ ..... **[3]**

(b) Point C lies on the x-axis.
The triangle ABC is isosceles with $AC = BC$.
Calculate the x-coordinate of C.

$x =$ ..... **[4]**

1 Half of the chord length $= 24 \div 2 = 12$ **[M1]**

Uses Pythagoras' theorem: $r^2 = 5^2 + 12^2$ **[M1]**

13 **[A1]**

[Back to Question] *[Total Marks: 3]*

2 Use of Pythagoras' theorem: $(k-1)^2 + (19-4)^2 = 17^2$ **[M1]**

$(k-1)^2 + 15^2 = 289$ or $(k-1)^2 = 64$ **[M1]**

$k - 1 = 8$ and $k - 1 = -8$ (or $k^2 - 2k - 63 = 0$

(k-9)(k+7) = 0)[$M1$]

9 and -7 **[A1]**

[Back to Question] *[Total Marks: 4]*

3 (a) $(x-7)^2 + x^2 = (x+1)^2$ **[M1]**

$x^2 - 14x + 49 + x^2 = x^2 + 2x + 1$ **[M1]**

$2x^2 - 14x + 49 = x^2 + 2x + 1$

$x^2 - 16x + 48 = 0$ **[A1]**

(b) $(x-4)(x-12) = 0$ **[M1]**

$x = 4$ or $x = 12$ **[A1]**

Rejects $x = 4$ because $x - 7$ would be -3 (length must be positive), so $x = 12$ **[B1]**

Perimeter $= 5 + 12 + 13 = 30$ **[A1]**

[Back to Question] *[Total Marks: 7]*

4 (a) $(2x+1)^2 = (x-1)^2 + (x+6)^2$ **[M1]**

$4x^2 + 4x + 1 = (x^2 - 2x + 1) + (x^2 + 12x + 36)$ **[M1]**

$4x^2 + 4x + 1 = 2x^2 + 10x + 37$ **[M1]**

$2x^2 - 6x - 36 = 0$

$x^2 - 3x - 18 = 0$ with no errors seen [A1]

(b) $(x-6)(x+3) = 0$ **[M1]**

$x = 6$ or $x = -3$ **[A1]**

Selects $x = 6$ and rejects $x = -3$ because lengths must be strictly positive **[B1]**

(c) Base $= 6 - 1 = 5$ and Height $= 6 + 6 = 12$ **[M1]**

Area $= \frac{1}{2} \times 5 \times 12$ **[M1]**

30 **[A1]**

[Back to Question] *[Total Marks: 10]*

5 **(a)** Let half the chord length be x. Uses $12.4^2 = x^2 + 5.2^2$ **[M1]**

$x = \sqrt{12.4^2 - 5.2^2}$ or $x = 11.257\ldots$ **[M1]**

22.5 (or 22.51 to 22.52) **[A1]**

(b) Let perpendicular distance from O to CD be y. Uses $12.4^2 = y^2 + 9^2$ **[M1]**

$y = \sqrt{12.4^2 - 9^2}$ or $y = 8.5299\ldots$ **[A1]**

Total distance $= 5.2 + 8.5299\ldots$ **[M1]**

13.7 (or 13.72 to 13.73) **[A1]**

[Back to Question] *[Total Marks: 7]*

6 **(a)** Uses distance formula: $\sqrt{(6-(-2))^2 + (y-(-1))^2} = 10$ **[M1]**

$64 + (y+1)^2 = 100$ **[M1]**

$(y+1)^2 = 36$

$y + 1 = 6$ (since $y > 0$)

5 **[A1]**

(b) Let C be $(x, 0)$.

$AC^2 = (x-(-2))^2 + (0-(-1))^2 = (x+2)^2 + 1$ **[M1]**

$BC^2 = (x-6)^2 + (0-5)^2 = (x-6)^2 + 25$ **[M1]**

Equating distances squared: $x^2 + 4x + 5 = x^2 - 12x + 61$ **[M1]**

$16x = 56$

3.5 **[A1]**

[Back to Question] *[Total Marks: 7]*